

Project Documentation

Group – 13

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Stock and Production Management System for Hamoud Boualem (Hespera)

Introduction

Hamoud Boualem is a historic Algerian beverage manufacturer, established in 1878. The company is celebrated for its high-quality traditional soft drinks, particularly its signature lemonade. Over the years, it has become a symbol of cultural heritage in Algeria. For more information about the company, visit their official website: <https://hamoud-boualem.com/en/>.

The "Hespera: Stock and Production Management System" is a tailored solution aimed at addressing the operational challenges faced by Hamoud Boualem. By streamlining stock and production processes, the system ensures efficient use of resources and enables real-time monitoring, ultimately enhancing productivity.

System Overview

The Stock and Production Management System is a software tool designed to optimize inventory control and production scheduling. The system allows for the addition, updating, and monitoring of stock items, provides alerts for low inventory, and supports batch production management. Key features include a user authentication mechanism and detailed report generation for informed decision-making. Built with object-oriented programming principles in C++.

Core Features

- **Inventory Management:**
 - Add, view, and update raw materials and finished products.
 - Categorize items for better organization.
 - Save and retrieve stock records using file handling.
- **Production Scheduling:**
 - Validate raw material availability before scheduling production.
 - Automatically deduct materials from stock after production.
- **Notifications:**
 - Alert users when stock levels fall below reorder thresholds.
- **Reports:**
 - Generate comprehensive stock and production summaries.

- Save reports externally for review and record-keeping.
- **Security:**
 - Secure login system with account lockout after multiple failed attempts.

Implementation Highlights

The system leverages a console-based interface for simplicity, with intuitive menu-driven navigation. Key technical aspects include:

- **Dynamic Memory Management:** Efficient use of memory through dynamically allocated arrays, allowing seamless scaling with increased stock items.
- **Object-Oriented Design:** The system use object-oriented programming principles, including inheritance to reuse and extend base class functionality, polymorphism for flexible handling of different data types, and abstraction to hide implementation details. This modular approach ensures efficient management of stock and production processes.
- **File Handling:** Persistent storage for stock data and reports, enabling data retrieval and analysis.

How to Use

1. **Login:** Use the default credentials (username: admin, password: 12345). The system locks after two unsuccessful attempts.
2. **Navigate:** Access the main menu for inventory and production management options.
3. **Save Data:** Use file-saving features to preserve stock records and reports for future reference.

That's all for now.

<https://hamoud-boualem.com/en/>.

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